[redacted]

$p_{Kein} = p_{Kaus} = konst$

1) $T_{ein} = 70°C$ $\quad T_{Kein} = 298.15\,K$ $\quad$ Wasser TABs

$T_{aus} = 100°C$ $\quad T_{Kaus} = 288.15\,K$ $\quad$ Kühl: perf. fluid

$T_{Reaut} = 100°C$

$m_{ges} = 5755\,kg$

$x_D = 0.005 = m_D/m_{ges}$

$\dot{Q}_R = 100\,kW$

$\frac{V}{\dot{V}} = m$

$\frac{1}{\rho} = v$

$w\frac{A}{v} = \dot{m}$

$\frac{\Delta H}{\dot{Q}} = \Delta t$

a) $\dot{Q}_{aus} = ?$

$$\left[\begin{matrix}\text{stat.}\\ 2\times\dot{m}\end{matrix}\right] \quad 0 = \dot{m}_{ein}(h_e - h_a) + \dot{m}_{aus}(h_e - h_a) + \dot{Q}_R + \dot{Q}_{aus} - \cancel{\sum \dot{W}}^{=0}$$

$$\curvearrowright \dot{Q}_{aus} = 2\dot{m}_{ein}(h_e - h_a) + \dot{Q}_R$$

$\leftarrow h_e = h_g(70°C) = 292.98\,\frac{kJ}{kg}$ $\quad$ 2626.8

$h_a = h_g(100°C) = 419.04\,\frac{kJ}{kg}$ $\quad$ 2676.1

TAB A2

$$\dot{Q}_{aus} = 2\cdot 0.3\,kg/s\,(292.98 - 419.04)\,\frac{kJ}{kg} + \dot{Q}\ 100\,kW$$

$$= \underline{\underline{24.364\,kW}}$$

b) $$\bar{T}_{KF} = \frac{\int_e^a T\,ds}{s_e - s_a}$$

$$\longrightarrow s_e - s_a = \int_a^e c\,\frac{1}{T}\,dT = c\cdot\ln(T_e/T_a)$$

$$\longrightarrow \int_e^a T\,ds = \Delta H = h_e - h_a = \int_a^e c\,dT + v^{i3}\cancel{(p_e - p_a)}^{=0\ \text{konstant}} = c(T_e - T_a)$$

$$\longrightarrow \frac{\cancel{c}\cdot T_e - T_a}{\cancel{c}\ln(T_e/T_a)} = \bar{T}_{KF} \quad |:c$$

$\left\{\begin{matrix} T_{Kein} = 288.15 \\ T_{Kaus} = 298.15\end{matrix}\right.$

$$\frac{288.15 - 298.15}{\ln(288.15/298.15)} = \underline{\underline{293.12\,K = \bar{T}_{KF}}}$$

c) $\dot{S}_{erz} = ?$

$$\left.\begin{matrix}\text{stat.}\\ 2\times\dot{m}\end{matrix}\right\} \quad 0 = \dot{m}_{ein}(s_e - s_a) + \dot{m}_{aus}(s_e - s_a) + \dot{Q}_{aus}/\bar{T}_{KW} + \dot{S}_{erz}$$

$$\curvearrowright \dot{S}_{erz} = 2\dot{m}_{ein}(s_a - s_e) - (\dot{Q}_{aus}/\bar{T}_{KW})$$

$\leftarrow s_e = s_g(15°) = 0.2245\,\frac{kJ}{kgK}$ $\quad$ 8.7614

$s_a = s_g(25°) = 0.367\,\frac{kJ}{kgK}$ $\quad$ 8.6580

$$= 2\cdot 0.3\,kg/s\,(0.367 - 0.2245)\,\frac{kJ}{kgK} - \frac{65\,kW}{295\,K}$$

$\frac{\dot{Q}_{aus}}{\bar{T}}$ } vorgegebene Werte

$$= \underline{\underline{-0.356\,kW/K}}$$

d) T

1 20°C

2 70°C

Δm_{12}

$Q_{aus} = 35 \cdot 10^3 \, kJ$

Zufluss gestoppt

d) $$\frac{dE}{dt} = \cancel{\dot{m}_{12}(h_1 - h_2)} + \sum \dot{Q} - \cancel{\sum \dot{W}_t} = 0$$

$$m_2 u_2 - m_1 u_1 = \cancel{\dot{m}_{12}(h_1 - h_2)} + \dot{Q}_{aus}$$

m_2

$\leftarrow$ $h_1 = h_g(20°C) = 83.96$

$h_2 = h_g(70°C) = 292.98$

$u_2 = u_g(70°C) = 1.0228 \cdot 10^{-3}$

$u_1 = u_g(20°C) = 1.0018 \cdot 10^{-3}$

[redacted]

$w_5 = 220\ m/s$

2) $w_a = 200\ m/s$

$\dot{m}_{ges}$

$\eta_{V,s} < 1 \leftarrow$ isentrop

$\frac{\dot{m}_H}{\dot{m}_K} = 5.293$

$q_B = \frac{\dot{Q}_B}{\dot{m}_K} = 1195\ \frac{kJ}{kg}$

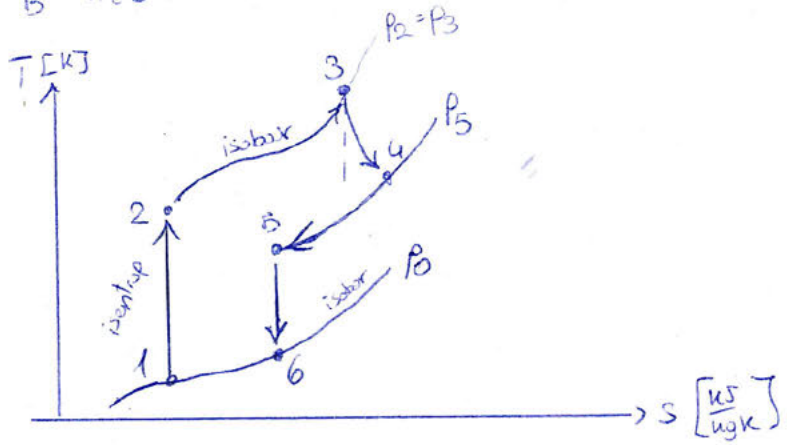
$\bar{T}_B = 1289$

	p	T
0	0.191 bar	−30°C
5	0.5 bar	431.9 K

$c_p = 1.006$

$n = \kappa = 1.4$

a)



b) $w_6 = ?\quad T_6 = ?$

$\cancel{w_{06}^{rev} = \int_0^6 p\,dv = R}$

$\cancel{\frac{T_6}{T_0} =} \frac{T_6}{T_5} = \left(\frac{p_6}{p_5}\right)^{\frac{n-1}{n}}$ ← isentrop $n = \kappa = 1.4$

$\curvearrowright \underline{T_6} = T_5 \left(\frac{p_0}{p_5}\right)^{\frac{n-1}{n}}$

$= 431.9\,K \left(\frac{0.191\ bar}{0.5\ bar}\right)^{\frac{0.4}{1.4}} = \underline{\underline{328.075\,K}} = T_6$

stat. $1 \times \dot{m}$ $\Big\}$ $0 = \dot{m}\left(h_e - h_a + \frac{w_e^2 - w_a^2}{2}\right) + \sum \dot{Q} - \sum \dot{W}_t$

c) $\dot{m}_{ges} = ?$

$\Delta e_{x\,stc} = ?$

$$\dot{m}_{ges}\left(h_0 - h_6 + \frac{w_2^2 - w_0^2}{2}\right) + \overset{=0}{\cancel{\sum \dot{Q}}} + \sum W_t \quad \leftarrow \quad W_t = \frac{\dot{W}_t}{\dot{m}} = \int_0^6 v\,dp + \Delta ke + \overset{=0}{\cancel{\Delta pe}}$$

$$= -n \int p\,dV + \Delta ke$$

$$= -n \frac{R(T_6 - T_0)}{1-n} + \frac{1}{2}\dot{m}(w_2 - w_0)^2$$

$$\dot{m}_{ges}\left(h_6 - h_0 - T_0(s_6 - s_0) + \Delta ke + \overset{=0}{\cancel{\Delta pe}}\right) = \Delta e_{x,stc}.$$

d) $ex_{verl} = T_0 \dot{S}_{erz}$

$= \cancel{T_0}$

$$\longrightarrow \dot{S}_{erz} = \dot{m}_{ges}(s_6 - s_0) - \overset{=0 \text{ adiabat}}{\cancel{\frac{\dot{Q}}{T}}}$$

[redacted]

3) a) $p_{G1} = ?$ $m_g =$

$A = (D/2)^2 \pi = 0.007854\ m^2$

$p_{neu} = F/A = m_u g/A + m_{EW} g/A$

$= 32 \cdot 9.81 / 0.007854 + 0.1\ kg\ 9.81 / 0.007854$

$= 40'009.35\ Pa$

$= 0.4\ bar$

$p_{G1} = p_0 + p_{neu} = \underline{\underline{1.4\ bar}}$

$m_g = ?$

$pV = mRT \Rightarrow m = \frac{pV}{RT} =$ $\qquad R = \frac{\bar{R}}{M} = \frac{8.314}{50} = 0.16628$

$p = p_{G1}$

$m_g = \frac{p_{G1} V}{RT} = \frac{1.4\ \cancel{bar} \cdot 10^2\ kPa \cdot 3.14\ L}{0.16628 \frac{kJ}{kgK} \cdot (500 + 273.15)\ K \cdot 1000}$

$= 0.00342\ kg$

$= \underline{\underline{3.42\ g}}$

b) Wenn $x = 0$ dann ist Eis = Wasser

Wenn $x = 1$ dann ist Eis = Dampf

$\rightarrow T_2$ muss kleiner werden

c) $\frac{dE}{dt} = \sum \dot{Q} - \sum \dot{W}$

$\Delta E = \Delta Q - \cancel{\Delta W}$ keine Arbeit

~~$m_2 = m_1$~~ verdoppelt

$m_2 u_2 - m_1 u_1 + \cancel{\Delta KE} + \cancel{\Delta PE} = \Delta Q$ $\qquad u_1 (500°C, 1.4\ bar)$

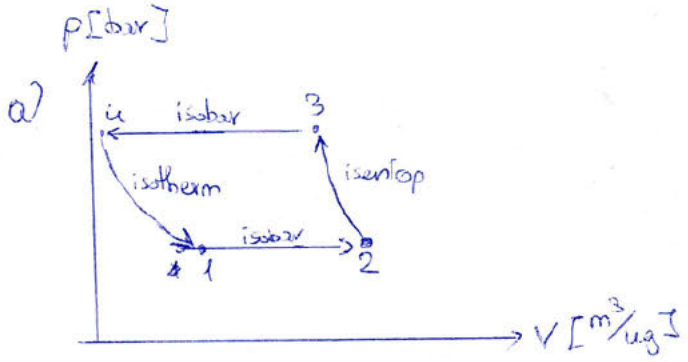
$u_2 (\quad , 0.003°C)$

[redacted]

a) R134a

	p	T			
1		$\boxed{31.33°C}$			
2			$\boxed{s_2 = s_3}$ $x_2 = 1$	gas	← isentrop
3	$\boxed{8\,bar}$				← $\cancel{\dot{W}_u = 28\,k}$ isobar
4	$\boxed{8\,bar}$	$\boxed{31.33°C}$	$x_4 = 0$	flüssig	← isenthalp + isotherm

$T_1 - 6k = T_2$



$T_1 = 10k + T_{Kappel}$
$= 10k + 273.15$
$= 283.15$

$T_2 = T_1 - 6k = 277.15$
$=$

b) $\dot{m} = ?$

stat. $1 \times \dot{m}$ } $0 = \dot{m}(h_e - h_a) + \sum \dot{Q} - \sum \dot{W}$
$=$

~~P2=P~~ $s_2 = s_g(6k) =$
$s_4 = s_g(\quad = 0.3459$
$T_4 = 31.33°C$
$s_3 = s_g(\cancel{8\,bar}) = 0.9066$
$h_1 = h_4 = h(31.33°C,$

c) $\dot{m}_R \cdot T_2 =$

$x_1 = \dfrac{h_1 - h_f}{h_g - h_f}$ $\longrightarrow T_1 = T_4 = 31.33°C$

d) $\varepsilon_k = \dfrac{\dot{Q}_{zu}}{\dot{W}_t} = \dfrac{|\dot{Q}_{zu}|}{|\dot{Q}_{ab}| - |\dot{Q}_{zu}|} =$

← $h_1 = h(T_1, p_1)$

$\longrightarrow \dot{Q}_{zu} = \dot{Q}_{ik} = \dot{m}(h_2 - h_1)$ ← $h_2 = h_g(T_2 = -22°C) = 234.08$

$\longrightarrow \dot{Q}_{ab} = \dot{m}(h_4 - h_3)$ ← $h_3(8\,bar, T_3)$

← $h_4 = h_f(8\,bar) = 93.42$

d) T_i würde immer niedriger werden
bis es "unmöglicherweise" zum absoluten
Nullpunkt erreicht